Budgets

Overview

CampusCents is a personal finance application designed to help college students manage expenses, track transactions, create budgets, and receive AI-powered financial recommendations. Two service-level objectives (SLOs) have been established: AI analysis response time and financial calculation correctness.

Error Budget Calculations

Target 1: AI Analysis Latency

Service-Level Objective (SLO): At least 95% of AI-powered expense categorization and financial insight requests must complete and display results within 3 seconds over a rolling 30-day period.

Error Budget Calculation: Error Budget = 100% − 95% = 5%

For example, if CampusCents processes 10,000 AI requests during the measurement period:

10,000 × 0.05 = 500 allowable slow requests.

Therefore, up to 500 requests may exceed the 3-second response-time target without violating the SLO.

Error Budget Policy: If more than 50% of the error budget is consumed, the development team will investigate slow API responses, AI processing delays, and frontend rendering performance. If the entire budget is exhausted, nonessential feature releases will be temporarily paused until performance improvements are implemented and verified.

Target 2: Financial Arithmetic Correctness

Service-Level Objective (SLO): CampusCents must maintain 100% accuracy in financial calculations and transaction synchronization at every sprint deployment deadline.

Error Budget Calculation:

Error Budget = 100% − 100% = 0%

This establishes a zero-tolerance policy for incorrect budget totals, transaction discrepancies, and synchronization failures.

Error Budget Policy: Any confirmed financial calculation error or corrupted transaction record immediately exhausts the error budget. The team must block or roll back the affected deployment, investigate the issue, correct the defect, and complete regression testing before releasing the application again.